

Gaussian domination of the fluctuation function:

$$S_\lambda(a) \leq e^{\beta a^2/2} (\beta/2)^{-\lambda} \Gamma(\lambda)$$

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2026-09-06

Abstract

The analytic backbone of grammar §4.2's tail and convergence estimates, exposed as reusable lemmas: the complete-the-square domination $e^{-\beta t + \beta a \sqrt{t}} \leq e^{-(\beta/2)t + \beta a^2/2}$ and the closed-form Gaussian upper bound $S_\lambda(a) \leq e^{\beta a^2/2} (\beta/2)^{-\lambda} \Gamma(\lambda)$ for $\lambda > 0$. Zero `sorry/axiom`.

1 Setting

Every boundary-remainder and convergence bound in §4.2 (culminating in `eq:Deltaupper` $|\Delta| \leq C e^{-\varepsilon n}$) begins with one pointwise inequality: the fluctuation exponent is sub-Gaussian in $\sqrt{t}$. Completing the square,

$$-\beta t + \beta a \sqrt{t} = -\frac{\beta}{2}t - \frac{\beta}{2}(\sqrt{t} - a)^2 + \frac{\beta}{2}a^2 \leq -\frac{\beta}{2}t + \frac{\beta}{2}a^2,$$

so the integrand of $S_\lambda(a) = \int_0^\infty t^{\lambda-1} e^{-\beta t + \beta a \sqrt{t}} dt$ is dominated by $e^{\beta a^2/2} t^{\lambda-1} e^{-(\beta/2)t}$, whose integral is an elementary Gamma integral. This domination was already buried inside `fluctuation_integrable0n`'s proof; here it is factored out and turned into a quotable bound.

2 The things formalised

```
theorem fluctuation_integrand_le ( a t : ) (h : 0 < ) (ht : 0 < t) :  
  Real.exp (- *t + *a*Real.sqrt t) Real.exp (-( /2)*t + *a^2/2)  
  
theorem fluctuation_le_gaussian ( a lam : ) (h : 0 < ) (hlam : 0 < lam) :  
  fluctuation lam a Real.exp (*a^2/2) * (/2)^(-lam) * Real.Gamma lam
```

3 Proof strategy

The pointwise lemma is `Real.exp_le_exp` composed with `nlinarith`, fed the two facts $(\sqrt{t})^2 = t$ (`Real.sq_sqrt`) and $\beta(\sqrt{t} - a)^2 \geq 0$ (`mul_nonneg` of `h.le` and `sq_nonneg`); `nlinarith` finds the square completion. The bound then follows by `setIntegral_mono_on` against the majorant $e^{\beta a^2/2} t^{\lambda-1} e^{-(\beta/2)t}$ — integrable by `integrable0n_rpow_mul_exp_neg_mul_rpow` at $p = 1$ — and evaluating the majorant integral by `integral_rpow_mul_exp_neg_mul_rpow`, which gives $(\beta/2)^{-\lambda} \Gamma(\lambda)$ at $p = 1$, $q = \lambda - 1$, $b = \beta/2$.

4 Roadblocks and resolutions

- The Gamma-integral lemmas are stated with the `rpow` form t^1 in the exponential; align the majorant integrand with $t^{(1:\mathbb{R})}$ and peel it with `Real.rpow_one` inside the pointwise comparison only.
- The lemma output carries `/1` divisions and the exponent $(\lambda - 1) + 1$; `simp only [div_one]` clears the former and `rw [show lam - 1 + 1 = lam by ring]` rewrites the exponent in both the `rpow` power and the Γ argument at once (a bare `ring` cannot reach under `rpow`/ Γ).
- Splitting $e^{-(\beta/2)t + \beta a^2/2} = e^{\beta a^2/2} e^{-(\beta/2)t}$ needs `Real.exp_add` then `ring` to reassociate the constant to the front, matching the majorant's shape.

5 What was Mathlib, what was new

Mathlib supplied the Gamma-integral pair `integrableOn_rpow_mul_exp_neg_mul_rpow` and `integral_rpow_mul_exp_neg_mul_rpow`, together with `setIntegral_mono_on`, `Real.sq_sqrt`, `Real.exp_le_exp`. New is the exposure of the domination as a standalone lemma and the closed-form fluctuation bound — neither was previously stated, though the majorant recurred inside the integrability proof.

6 Lessons

When an integrability proof already constructs a majorant, its pointwise domination and the value of the majorant integral are two quotable lemmas waiting to be factored out; do so when downstream estimates (here the §4.2 remainder) will lean on the bound repeatedly.

7 Follow-ups

The bound is the first ingredient of the boundary-tail remainder Δ (`eq:Deltaupper`); the remainder itself additionally needs the Taylor-tree coefficient bounds (`prop:CoefficientBound`) and is multivariate.